

Chapter 9: Force and Laws of Motion

Class 9 Science — Physics | Greenfield Public School, Pune | Study Notes 2026-27

9.1 Balanced and Unbalanced Forces

A force is a push or a pull that can change the state of rest or of uniform motion of an object, change its speed, change its direction, or change its shape. Force is measured in newtons (N).

When two or more forces act on an object and their combined effect is zero, the forces are called balanced forces. Balanced forces do not change the state of rest or motion of the object. During our class activity on 14 July, two students, Aarav and Meher, pulled a wooden block from opposite sides with spring balances that both read 12 N. The block did not move, because the two 12 N forces were balanced.

When Meher's spring balance was increased to 18 N while Aarav stayed at 12 N, the block slid towards Meher. The resultant, or net, force was $18\text{ N} - 12\text{ N} = 6\text{ N}$ towards Meher. Forces that produce a non-zero net force are called unbalanced forces, and only an unbalanced force can change the velocity of an object.

Friction is a force that always acts opposite to the direction of motion between two surfaces in contact. A football rolling on the school ground stops after some time because the friction between the ball and the grass is an unbalanced force acting against its motion.

9.2 First Law of Motion and Inertia

Newton's first law of motion states: an object remains in a state of rest or of uniform motion in a straight line unless compelled to change that state by an applied unbalanced force.

The natural tendency of an object to resist a change in its state of rest or of motion is called inertia. The mass of an object is a measure of its inertia: a heavier object has more inertia than a lighter one.

Everyday examples of inertia: (a) When a Pune city bus starts suddenly, standing passengers fall backwards, because their feet move with the bus while the upper body tends to stay at rest. (b) When the bus brakes suddenly, passengers lean forward, because their bodies tend to keep moving. (c) Dust comes out of a carpet when it is beaten with a stick, because the carpet moves but the dust tends to remain at rest.

Coin-and-card activity: place a playing card on an empty glass and a one-rupee coin on the card. Flick the card sharply with a finger. The card flies off, but the coin drops into the glass, because the coin's inertia keeps it at rest while the card moves away.

Seat belts are provided in cars so that when a car stops suddenly, the belt applies a force on the passenger that slows the body down gradually instead of letting it continue forward into the dashboard.

9.3 Momentum

The momentum (p) of an object is the product of its mass (m) and its velocity (v): $p = m \times v$. Momentum has both magnitude and direction, and its SI unit is kilogram metre per second (kg m/s).

Worked example 9.3: A cricket ball of mass 160 g is bowled by Riya at a speed of 25 m/s. Its momentum is $p = 0.16 \text{ kg} \times 25 \text{ m/s} = 4 \text{ kg m/s}$.

A slow-moving loaded truck can have far more momentum than a fast cricket ball, because its mass is thousands of times larger. This is why a truck is much harder to stop than a bicycle moving at the same speed.

9.4 Second Law of Motion

Newton's second law of motion states that the rate of change of momentum of an object is proportional to the applied unbalanced force, in the direction of the force. For an object of constant mass this gives $F = m \times a$, where a is the acceleration. One newton is the force that gives a mass of 1 kg an acceleration of 1 m/s^2 .

Worked example 9.4: The school's hand trolley has a mass of 40 kg. Ramesh bhaiya, the lab assistant, pushes it with a net force of 60 N. Its acceleration is $a = F / m = 60 \text{ N} / 40 \text{ kg} = 1.5 \text{ m/s}^2$.

The second law also explains why a fielder pulls his hands backwards while catching a fast ball. Moving the hands back increases the time taken to stop the ball, which reduces the rate of change of momentum and therefore reduces the force on the hands. For the same reason, high jumpers land on a thick foam bed or on a heap of sand.

9.5 Third Law of Motion

Newton's third law of motion states that to every action there is an equal and opposite reaction. The two forces always act on two different objects, so they never cancel each other out.

Examples: (a) When a gun is fired, it recoils backwards as the bullet moves forward. (b) A swimmer pushes the water backwards and the water pushes the swimmer forward. (c) When a sailor jumps out of a boat onto the ghat, the boat moves backwards into the river. (d) A rocket pushes hot exhaust gases downwards, and the gases push the rocket upwards.

Class demonstration: two students, each standing on a skateboard, held the ends of a rope. When only Kabir pulled the rope, both skateboards moved towards each other, showing that the force Kabir applied on the rope was matched by an equal and opposite force on him.

9.6 Conservation of Momentum

When two objects collide, and no external unbalanced force acts on them, the total momentum before the collision equals the total momentum after the collision. This is the law of conservation of momentum.

Worked example 9.6: A toy gun of mass 1.2 kg fires a rubber pellet of mass 20 g with a

velocity of 30 m/s. Before firing, the total momentum is zero. After firing, the pellet's momentum is $0.02 \text{ kg} \times 30 \text{ m/s} = 0.6 \text{ kg m/s}$ forward, so the gun must recoil with 0.6 kg m/s backwards. The recoil velocity of the gun is $0.6 / 1.2 = 0.5 \text{ m/s}$.

Quick Revision and Common Mistakes

1. Balanced forces cannot change motion; only an unbalanced (net) force can.
2. Inertia depends only on mass, not on speed. A parked truck and a moving truck of the same mass have the same inertia.
3. Action and reaction act on different objects, so they never cancel. A common mistake is to say the horse cannot pull the cart because the forces cancel.
4. Momentum is a vector: its direction matters. In the toy gun example the total momentum stays zero only because the pellet and the gun move in opposite directions.
5. Units to remember: force in newtons (N), momentum in kg m/s, acceleration in m/s^2 .